NORTHERN ARIZONA UNIVERSITY

School of Informatics, Computing, and Cyber Systems

Term	Class No.	Units	Days & Times	Room	Mode
Spring 2026	CS572	3	Tu Th 9.35-10:50 AM	SICCS – Rm 102	In person
Spring 2026	CS472	3	Tu Th 9.35-10:50 AM	SICCS – Rm 102	In person

Enrollment Requirements:

CS 470 with a grade of C or better

Basic Understanding of Machine Learning.

Suggested/soft pre-requisites: linear algebra, calculus, some experience programming in a high-level language with data visualization capabilities.

Instructor:

Dr. Khondker Fariha Hossain

Email: Khondker-Fariha.Hossain@nau.edu

Office Hours:

Day	Time	Location
Tuesday	12.00 A.M - 1.30 P.M	SICCS 222
Wednesday	12.00 A.M - 1.30 P.M	SICCS 222

Course Purpose

This course introduces advanced topics in machine learning with a focus on unsupervised and self-supervised methods. Unlike courses that primarily address supervised learning, where algorithms are trained on labeled data, this course emphasizes techniques for discovering structure in unlabeled data. Students will explore clustering, dimensionality reduction, change-point detection, and modern representation learning approaches.

Course Student Learning Outcomes:

- **Quizzes and Midterm (10%)**
  Assess student ability to:
  - Describe and explain foundational concepts in unsupervised and self-supervised learning (**LO1**)
  - Analyze problems and select appropriate algorithms (**LO2**)
  - Design pseudocode or workflows from models and optimization problems (**LO3**)
- **Weekly Assignments (class activity)**
  Assess student ability to:
  - Apply unsupervised learning methods to real-world datasets (**LO4**)
  - Evaluate algorithms using appropriate metrics and validation techniques (**LO5**)
- **Graduate Student Extensions**
  Additional exercises require students to:
  - Implement core algorithms from scratch (**LO6**)
  - Compare their implementations with open-source libraries (**LO7**)
- **Paper Presentations (30%)**
  Assess student ability to:
  - Synthesize and critique state-of-the-art research (**LO8**)
  - Communicate findings effectively in written and oral formats (**LO9**)
- **Final Project (40%)**
  Semester-long project assessing ability to:
  - Design, implement, and evaluate an unsupervised or self-supervised learning system end-to-end (**LO10**)
  - Graduate students must demonstrate additional **novelty or technical depth**.

Assessment Breakdown:

Component	Percentage
Participation & Quizzes	10%
Midterm Exam	20%
Group Presentation on SOTA Papers	30%
Semester-Long Final Project	40%

Grades will be assigned using the weighted sum described above using this scale:
A ≥ 90%, **B** ≥ 80%, **C** ≥ 70%, **D** ≥ 60%, **F** < 60%.

There is no "curve". Each student's grade is based on their own outcomes assessments and not affected by the grades of other students. Extra credit opportunities may present themselves throughout the semester and will be announced during class meetings. Mistakes in grading do happen, and students are encouraged to discuss such concerns with the instructor during office hours.

Texts:

Bishop – Pattern Recognition and Machine Learning
Murphy – Machine Learning: A Probabilistic Perspective
Hastie et al. – Elements of Statistical Learning
Weekly SOTA research papers (links provided on course site)

Class Outline and Tentative Schedule:

	Topics
Week 1	Introduction & Applications
Week 2 & 3	Clustering I: K-means & spectral methods
Week 3 &4	Clustering II: GMMs & EM algorithm
Week 5	Student Paper Presentations 1
Week 6	Hierarchical & Density-based Clustering
Week 7	Final Project Overview
Week 8	PCA & Probabilistic PCA.
Week 9	Spring Break
Week 10	Midterm
Week 11	Final Project update
Week 12	Student Paper Presentations 2
Week 13	Autoencoders
Week 14	Emerging Methods – Diffusion
Week 15	Self-Supervised Learning
Week 15	Final Project Presentation Day 1
Week 16	Final Project Presentation Day 2

Course Policies

The following policies will apply to this course:

- There will be no make-ups or late work accepted.
- There may be extra credit assignments given.
- **Cheating and plagiarism are strictly prohibited**. All work you submit for grading must be your own -- for the coding projects this means that you are not allowed to copy code that you found on the web, and submit that code as your own. The point of the coding projects is for you to take the time to learn how to code the machine learning algorithms and reproduce result figures from scratch. It is OK to discuss intellectual aspects with other students during the coding projects, but is it NOT OK to copy from other students, nor from other sources (e.g. code found on the internet). All academic integrity violations are treated seriously. Academic integrity violations will result in penalties including, but not limited to, a zero on the assignment, a failing grade in the class, or expulsion from NAU.
- Electronic device usage must support learning in the class. All cell phones, PDAs, music players and other entertainment devices must be turned off (or in silent mode) during lecture, and may not be used at any time. Laptops or workstations (if present) are allowed for note-taking and activities only during lectures; no web surfing or other use is allowed. I devote 100% of my attention to providing a high-quality lecture; please respect this by devoting 100% of your attention to listening and participating.
- Grades will be entered in BBLearn but your final grade will be calculated using the grading system described above and then entered in LOUIE. Your final course grade will **not** necessarily appear in BBLearn. Please check LOUIE for your final grade.
- Email to the instructor must be respectful and professional. Specifically, all emails should:
  - The body should contain complete sentences and correct grammar including correct usage of lowercase and uppercase letters. Composing emails on a mobile device is **not** an excuse for poor writing.
  - The body of your message should also be respectful and explain the full context of the query.
  - The subject should be prefixed with "CS472/572" so that the message can be easily identified or placed in an auto-folder. The subject should also use lower case and upper case correctly.
  - Although email will typically be answered quickly, you should allow up to three (3) business days for a response.
  - If you have a question that would require a long response or you have a lot of questions, please come to office hours or schedule an appointment with the instructor.
- Visiting the instructor(s) during office hours is encouraged! I am happy to talk about the class, careers, research, and topics related (even loosely) to this course.
- Anonymous feedback via the parking lot (put a sticky note on the door as you leave the classroom).

Appendix A. UNIVERSITY POLICY STATEMENTS

ACADEMIC INTEGRITY

NAU expects every student to firmly adhere to a strong ethical code of academic integrity in all their scholarly pursuits. The primary attributes of academic integrity are honesty, trustworthiness, fairness, and responsibility. As a student, you are expected to submit original work while giving proper credit to other people's ideas or contributions. Acting with academic integrity means completing your assignments independently while truthfully acknowledging all sources of information, or collaboration with others when appropriate. When you submit your work, you are implicitly declaring that the work is your own. Academic integrity is expected not only during formal coursework, but in all your relationships or interactions that are connected to the educational enterprise. All forms of academic deceit such as plagiarism, cheating, collusion, falsification or fabrication of results or records, permitting your work to be submitted by another, or inappropriately recycling your own work from one class to another, constitute academic misconduct that may result in serious disciplinary consequences. All students and faculty members are responsible for reporting suspected instances of academic misconduct. All students are encouraged to complete NAU's online academic integrity workshop available in the E-Learning Center and should review the full *Academic Integrity* policy available at https://www9.nau.edu/policies/Client/Details/1443?whoIsLooking=Students&pertainsTo=All

ARTIFICIAL INTELLIGENCE

Artificial intelligence (AI) technologies bring both opportunities and challenges. Ensuring honesty in academic work creates a culture of integrity and expectations of ethical behavior. The use of these technologies can depend on the instructional setting, varying by faculty member, program, course, and assignment. Please refer to course policies, any additional course-specific guidelines in the syllabus, or communicate with the instructor to understand expectations. NAU recognizes the role that these technologies will play in the current and future careers of our graduates and expects students to practice responsible and ethical use of AI technologies to assist with learning within the confines of course policies.

COPYRIGHT INFRINGEMENT

All lectures and course materials, including but not limited to exams, quizzes, study outlines, and similar materials are protected by copyright. These materials may not be shared, uploaded, distributed, reproduced, or publicly displayed without the express written permission of NAU. Sharing materials on websites such as Course Hero, Chegg, or related websites is considered copyright infringement subject to United States Copyright Law and a violation of NAU Student Code of Conduct. For additional information on ABOR policies relating to course materials, please refer to ABOR Policy 6-908 A(2)(5).

COURSE TIME COMMITMENT

Pursuant to Arizona Board of Regents guidance (ABOR Policy 2-224, *Academic Credit*), each unit of credit requires a minimum of 45 hours of work by students, including but not limited to, class time, preparation, homework, and studying. For example, for a 3-credit course a student should expect to work at least 8.5 hours each week in a 16-week session and a minimum of 33 hours per week for a 3-credit course in a 4-week session.

DISRUPTIVE BEHAVIOR

Membership in NAU's academic community entails a special obligation to maintain class environments that are conductive to learning, whether instruction is taking place in the classroom, a laboratory or clinical setting, during course-related fieldwork, or online. Students have the obligation to engage in the educational process in a manner that does not interfere with normal class activities or violate the rights of others. For additional information, see NAU's Student Code of Conduct policy at https://nau.edu/university-policy-library/student-code-of-conduct/.

NONDISCRIMINATION AND ANTI-HARASSMENT

NAU prohibits discrimination and harassment based on sex, gender, gender identity, race, color, age, national origin, religion, sexual orientation, disability, veteran status and genetic information. Certain consensual amorous or sexual relationships between faculty and students are also prohibited as set forth in the *Consensual Romantic and Sexual Relationships* policy. The Equity and Access Office (EAO) responds to complaints regarding discrimination and harassment that fall under NAU's *Nondiscrimination and Anti- Harassment* policy. To report a concern related to possible unlawful discrimination or harassment or to request a time to meet, please use the Report an Issue Form. To file a complaint, please submit the online Complaint Form. EAO also assists with religious accommodations. To request a religious accommodation, please use the Religious Accommodation Request Intake Form. EAO additionally provides access to lactation spaces, and please use to the Lactation Space Request Form to request use of a location. For additional information about nondiscrimination or anti-harassment, contact EAO at EquityandAccess@nau.edu, or visit the EAO website at https://nau.edu/equity-and-access. The EAO is located in Old Main on the first floor.

TITLE IX

Title IX of the Education Amendments of 1972, as amended, protects individuals from discrimination based on sex in any educational program or activity operated by recipients of federal financial assistance. In accordance with Title IX, Northern Arizona University prohibits discrimination based on sex or gender in all its programs or activities. Sex discrimination includes sexual harassment, sexual assault, relationship violence, and stalking. NAU does not discriminate on the basis of sex in the education programs or activities that it operates, including in admission and employment. NAU is committed to providing an environment free from discrimination based on sex or gender and provides a number of supportive measures that assist students, faculty and staff employees, and covered guests.

One may direct inquiries concerning the application of Title IX to either or both the university Title IX Coordinator or the U.S. Department of Education, Assistant Secretary, Office of Civil Rights. You may contact NAU's Title IX Coordinator at titleix@nau.edu or by phone at 928-523-5434 . In furtherance of its Title IX obligations, NAU promptly will investigate or equitably resolve all reports of sex/gender-based discrimination, harassment, or sexual misconduct and will eliminate any hostile environment as defined by law. To submit a report, please use the File a Report Form. The Office for the Resolution of Sexual Misconduct (ORSM): Title IX Institutional Compliance, Prevention & Response addresses matters that fall under the university's Sexual Misconduct Policy. ORSM also facilitates reasonable modifications for pregnant or parenting individuals. Additional important information and related resources, including how to request help or confidential support following conduct covered by the Sexual Misconduct Policy, is available on the ORSM web site, and you also may contact the office at titleix@nau.edu. The ORSM is located in Gammage on the third floor.

ACCESSIBILITY

Professional disability specialists are available at Disability Resources to facilitate a range of academic support services and accommodations for students with disabilities. If you have a documented disability, you can request assistance by contacting Disability Resources at 928-523-8773 (voice), ,928-523-8747 (fax), or dr@nau.edu (e-mail). Once eligibility has been determined, students register with Disability Resources every semester to activate their approved accommodations. Although a student may request an accommodation at any time, it is best to initiate the application process at least four weeks before a student wishes to receive an accommodation. Students may begin the accommodation process by submitting a self-identification form online or by contacting Disability Resources. The Director of Disability Resources, Jamie Axelrod, serves as NAU's Americans with Disabilities Act Coordinator and Section 504 Compliance Officer. He can be reached at jamie.axelrod@nau.edu

RESPONSIBLE CONDUCT OF RESEARCH

Students who engage in research at NAU must receive appropriate Responsible Conduct of Research (RCR) training. This instruction is designed to help ensure proper awareness and application of well-established professional norms and ethical principles related to the performance of all scientific research activities. More information regarding RCR training is available at https://legacy.nau.edu/university-policy-library/research/

MISCONDUCT IN RESEARCH

As noted, NAU expects every student to firmly adhere to a strong code of academic integrity in all their scholarly pursuits. This includes avoiding fabrication, falsification, or plagiarism when conducting research or reporting research results. Engaging in research misconduct may result in serious disciplinary consequences. Students must also report any suspected or actual instances of research misconduct of which they become aware. Allegations of research misconduct should be reported to your instructor or the University's Research Integrity Officer, Scott Pryor, who can be reached at scott.pryor@nau.edu or 928-523-5927. More information about misconduct in research is available at https://legacy.nau.edu/university-policy-library/research/

SENSITIVE COURSE MATERIALS

University education aims to expand student understanding and awareness. Thus, it necessarily involves engagement with a wide range of information, ideas, and creative representations. In their college studies, students can expect to encounter and to critically appraise materials that may differ from and perhaps challenge familiar understandings, ideas, and beliefs. Students are encouraged to discuss these matters with faculty.

Last revised August 14, 2025

School of Informatics, Computing, and Cyber Systems Initiatives

SICCS Freshman/Sophomore Academic Support
Student retention is a top priority for the Steve Sanghi College of Engineering (SCE) and the School of Informatics, Computing, and Cyber Systems (SICCS). To help you succeed, we provide a variety of academic resources, including office hours, tutoring, Peer Academic Coaching, and time management support.

If you are enrolled in a ***100-level or 200-level CS-, EE, SE, IMG, or CYB course****, you have access to FREE tutoring in Engineering Building Room 104. If you attend tutoring, you are eligible to receive a percentage of the points you missed on one early assignment or assessment in the course for which you sought tutoring. This grade adjustment is limited to one modification per course per semester and applies only to 100-level or 200-level CS-, EE, SE, IMG, or CYB prefixed course.*

Tutoring begins in Week 2. Schedules will be sent to faculty and students during Week 1 and posted in academic buildings. For more information, please contact Leslie Mitchell, our Academic Success Program Manager. *Leslie.Mitchell@nau.edu*

100% career readiness extra-credit opportunities
SCE and SICCS are committed to preparing students for their careers. We partner with industry and offer opportunities to engage with professionals through Industry Nights and the NAU Engineering Fest. To encourage participation, students can earn extra credit in ***CS-, EE-, SE-, IMG-, or CYB-prefixed courses*** *as follows:*

Each ***Industry Night*** *talk you attend allows you to add 2% to your final grade in one CS-, EE-, SE-, IMG-, or CYB course of your choice (in the same semester). You may apply at most 2% extra credit per course per semester from Industry Nights. Example: If you attend three Industry Nights, you may distribute the credits across courses, but no single course may receive more than 2% from Industry Night events. There are typically 4-6 Industry Night events throughout the semester and details about each event will be distributed separately through email and Canvas notifications*

By attending the ***NAU Engineering Fest****, you may add 3% to your final grade in one CS-, EE-, SE-, IMG-, or CYB course of your choice (in the same semester). This 3% is in addition to the 2% maximum from Industry Nights. Example: You may apply up to 5% total extra credit in one course (2% from Industry Nights + 3% from Engineering Fest). NAU Engineering FEST occurs every Fall and details about the event will be distributed through email and Canvas notifications.*

Last revised August 21, 2025